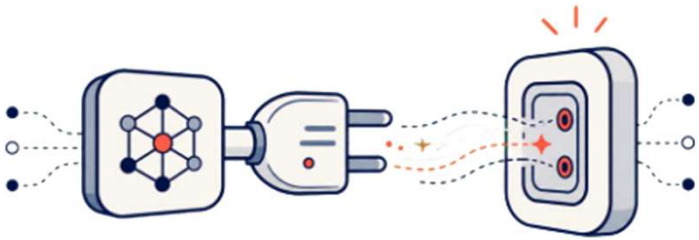


APIs

Muhammad Abbas

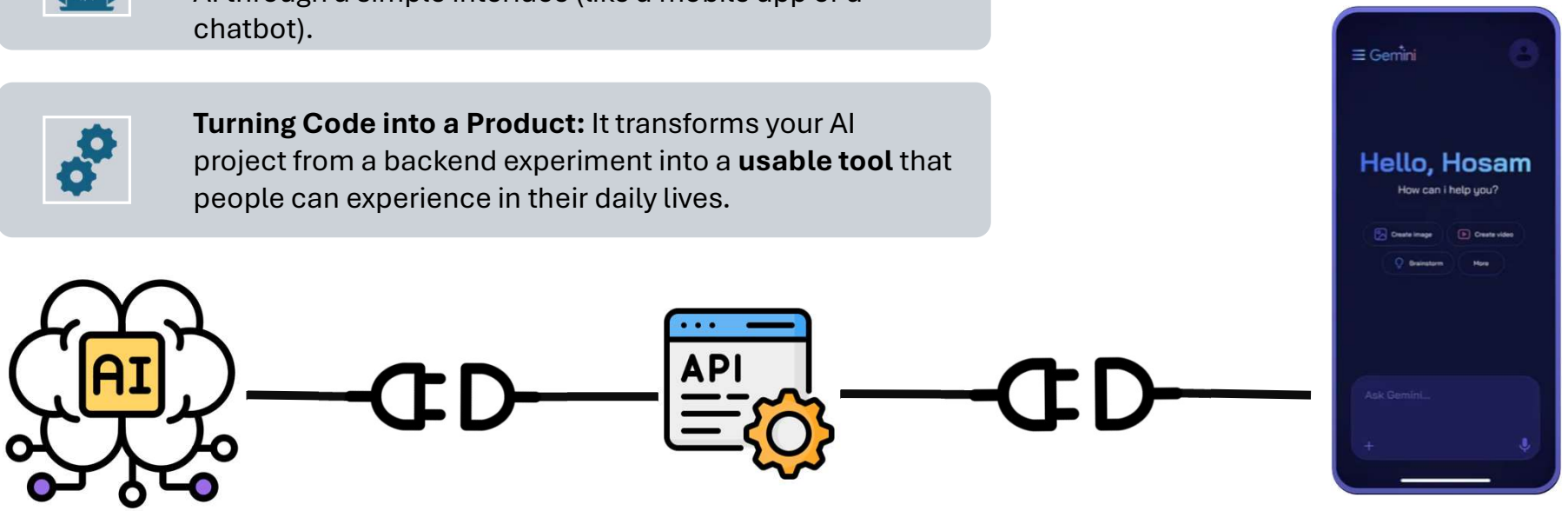
Giving Life to Your AI: Why We Need an API

The Bridge to the Real World: You trained an AI model, but it currently lives in isolation on your computer. An API is the bridge that lets it talk to websites, apps, and everyday devices.



Making AI Accessible: Users don't want to run Python code in the terminal. An API lets people interact with your AI through a simple interface (like a mobile app or a chatbot).

Turning Code into a Product: It transforms your AI project from a backend experiment into a **usable tool** that people can experience in their daily lives.



Your AI Model **Daily Life App**

What is an API?

APIs are mechanisms that enable two software components; such as the **frontend** and **backend** of and application; to **communicate** with each other using a define set of **rules**, **protocols**, and **data formats**.

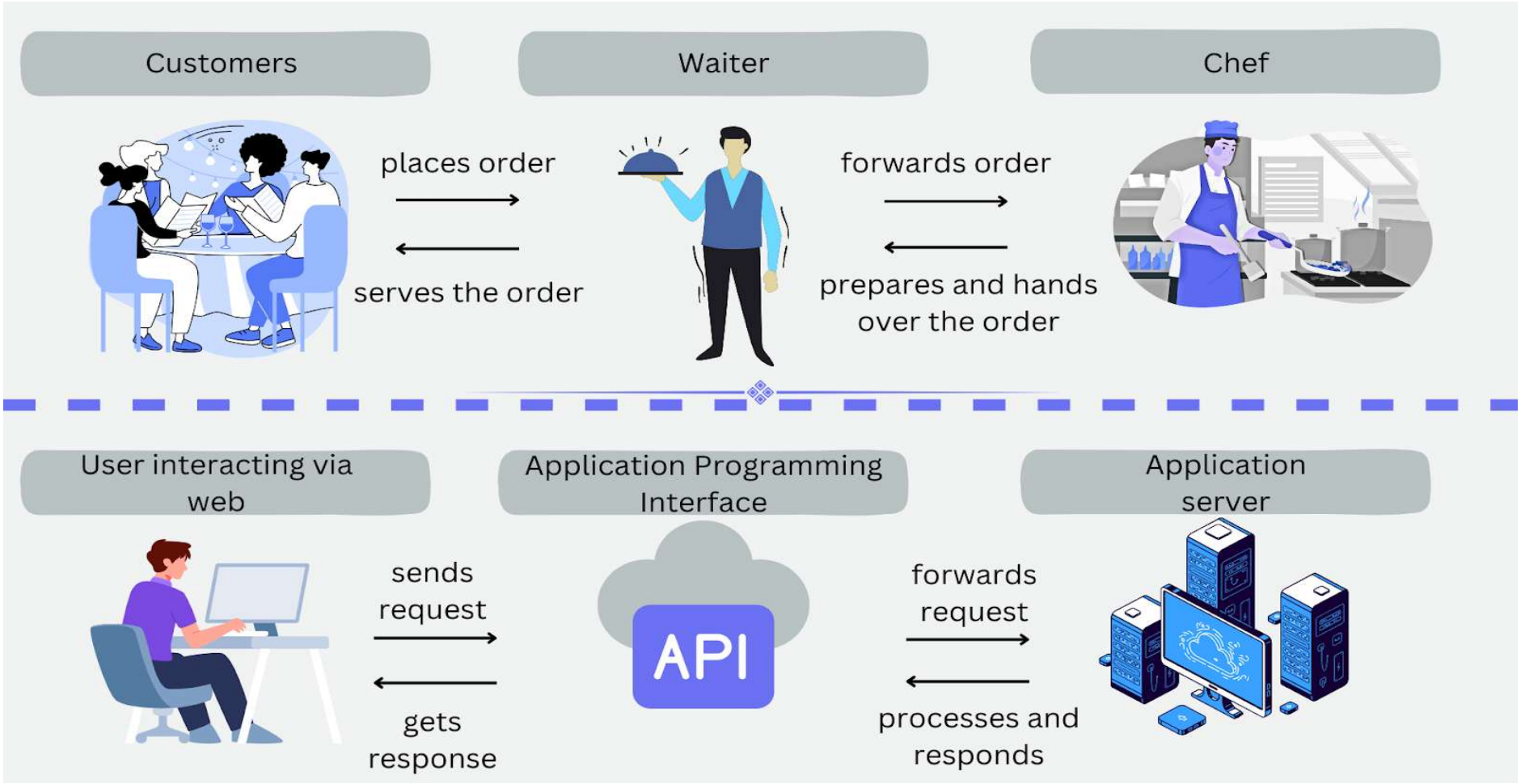
An Interface for Applications to Programmatically talk to one another.



FRONTEND

FRONTEND

The Restaurant Analogy

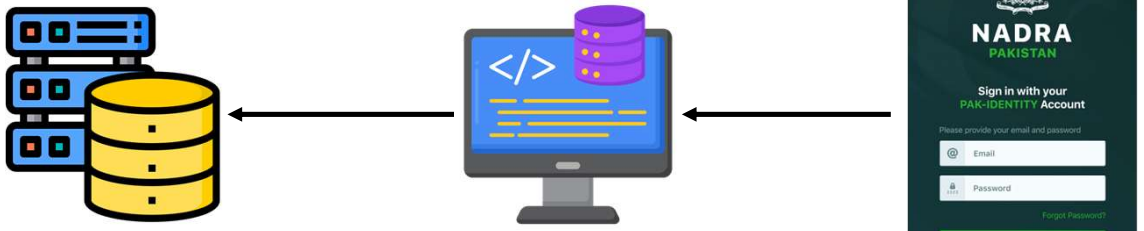


Need for APIs: The Monolithic Approach

The National Registry: NADRA maintains a massive, centralized database storing civil records and biometric data for over 240 million Pakistani citizens.

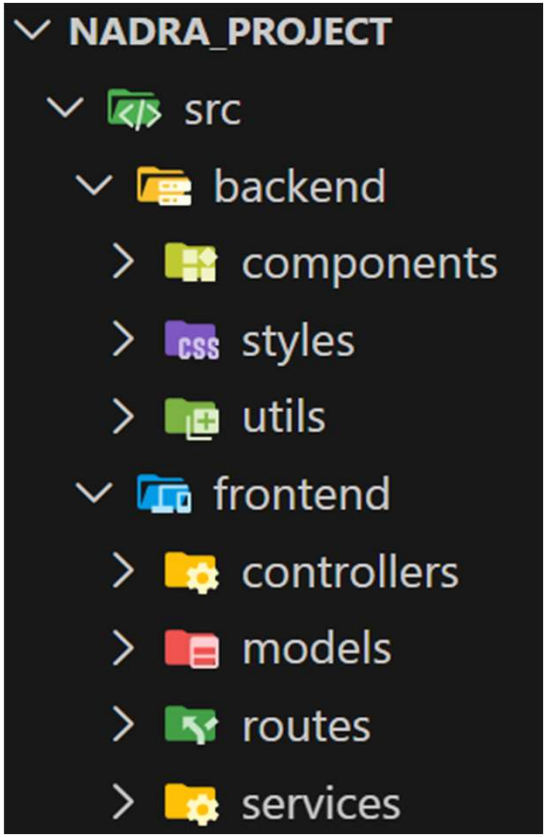
Internal Processing: In a monolithic setup, everything is tightly bundled. The user records, matching algorithms, and core system logic live inside a single, unified infrastructure block.

Direct Internal Access: Internal department or administrator runs a check; the code interacts directly with the database files within the same secure network.



Database **Backend** **Frontend**

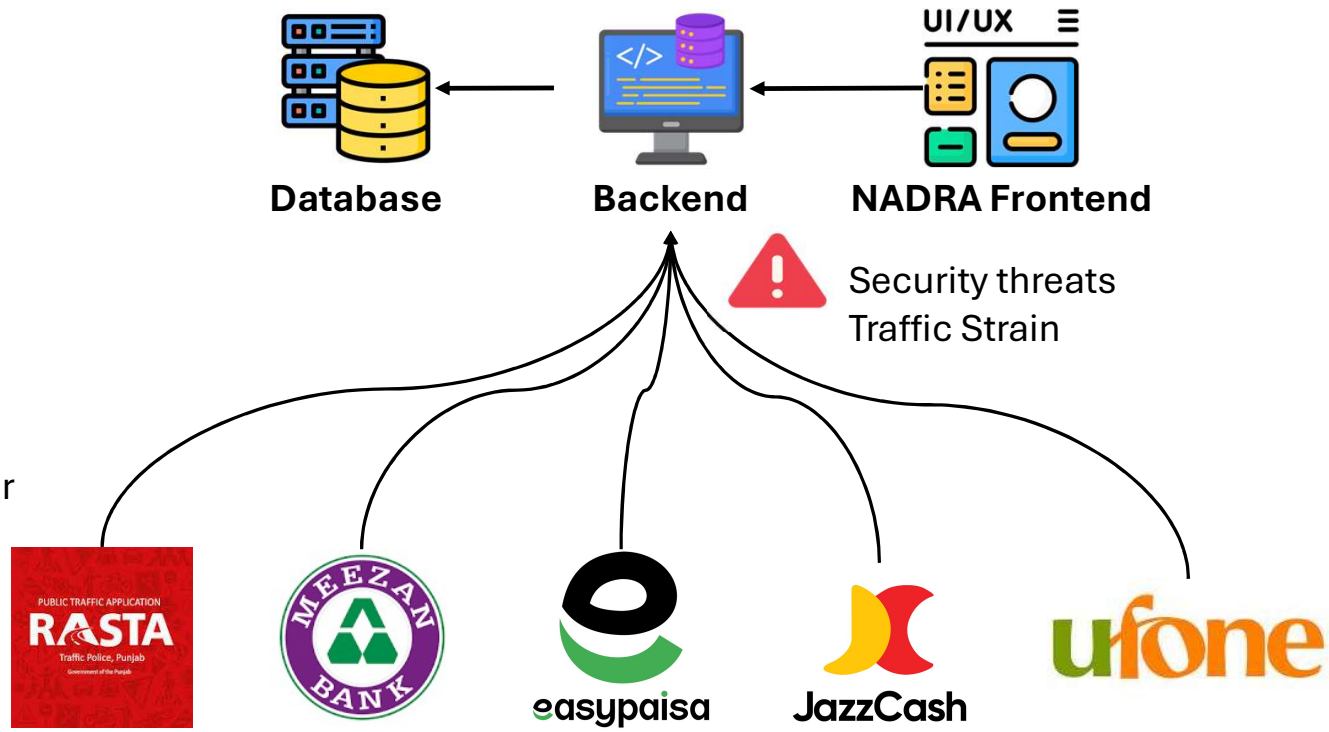
Monolithic Architecture



The Monolithic Dilemma: Connecting External Apps

The Multi-App Demand:
Fintechs and banks across Pakistan (like Easypaisa, NayaPay, and JS Bank) need to verify customer CNICs instantly to open accounts.

The Dangerous Solution (Direct Access):
If we stick to a monolithic mindset, we would have to give external apps direct access to our internal backend code and database files.

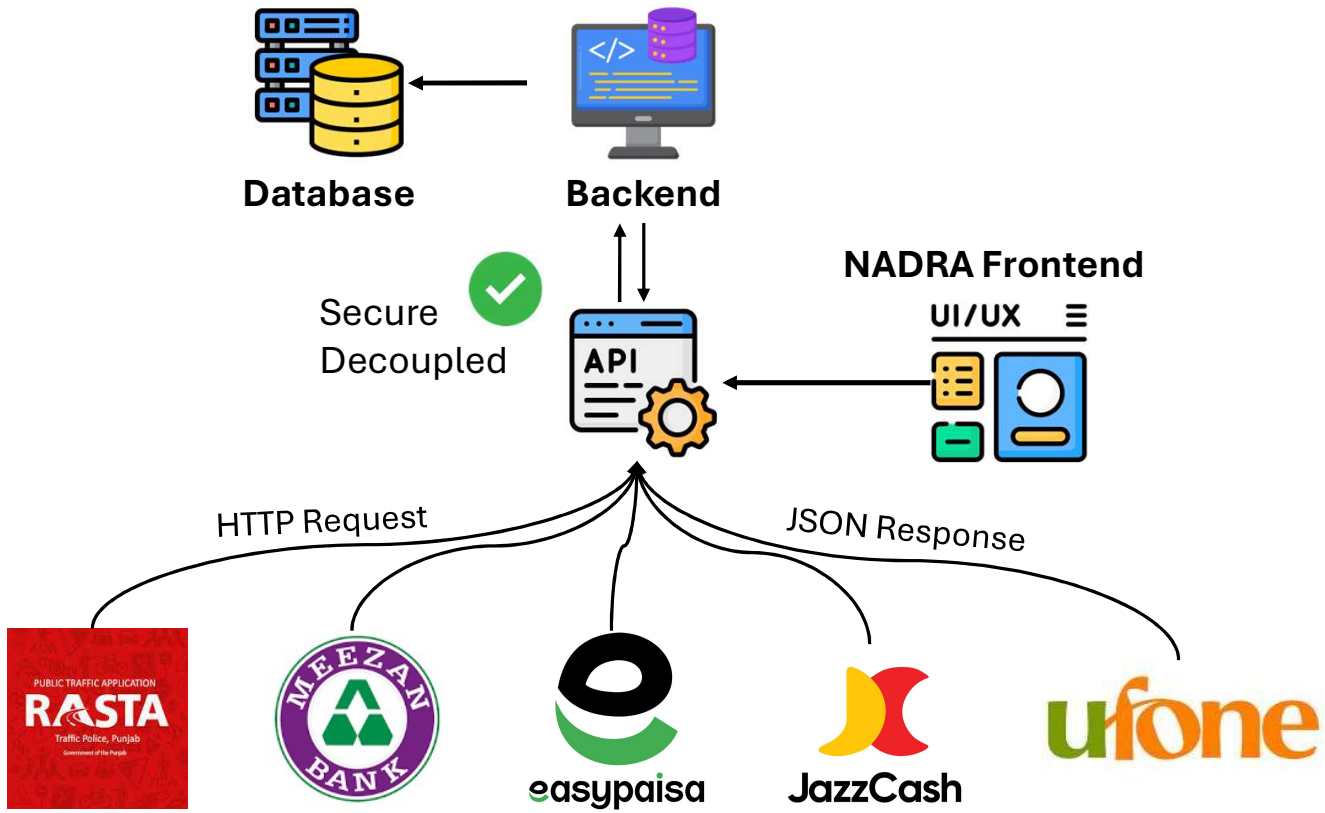


The API Solution: Secure Gateways

The Smart Middleware:
Instead of giving direct database access, we build an **API Gateway** (like NADRA's *Nishan Pakistan* portal).

Controlled Handshake:
Fintech apps send a structured request (CNIC + Fingerprint) via an API. The API checks the database securely and returns a simple answer (Verified or Rejected).

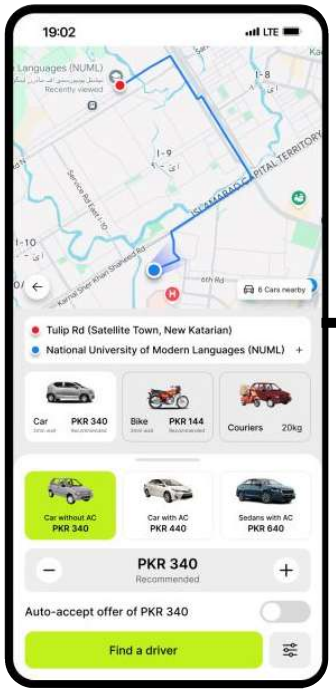
Multi-Platform Access:
APIs use a universal language (like JSON over HTTP), allowing different frontends, Android, iOS, or web, to easily connect to the same backend



Examples of APIs

Does inDrive launched
its own satellites





Request

Google Maps APIs



Google Data Center

Examples of APIs



Map and location APIs



Weather APIs



Client Application

API request



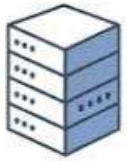




AI Model API



Google AI



Internal Models



Anthropic

AI model & generative AI APIs

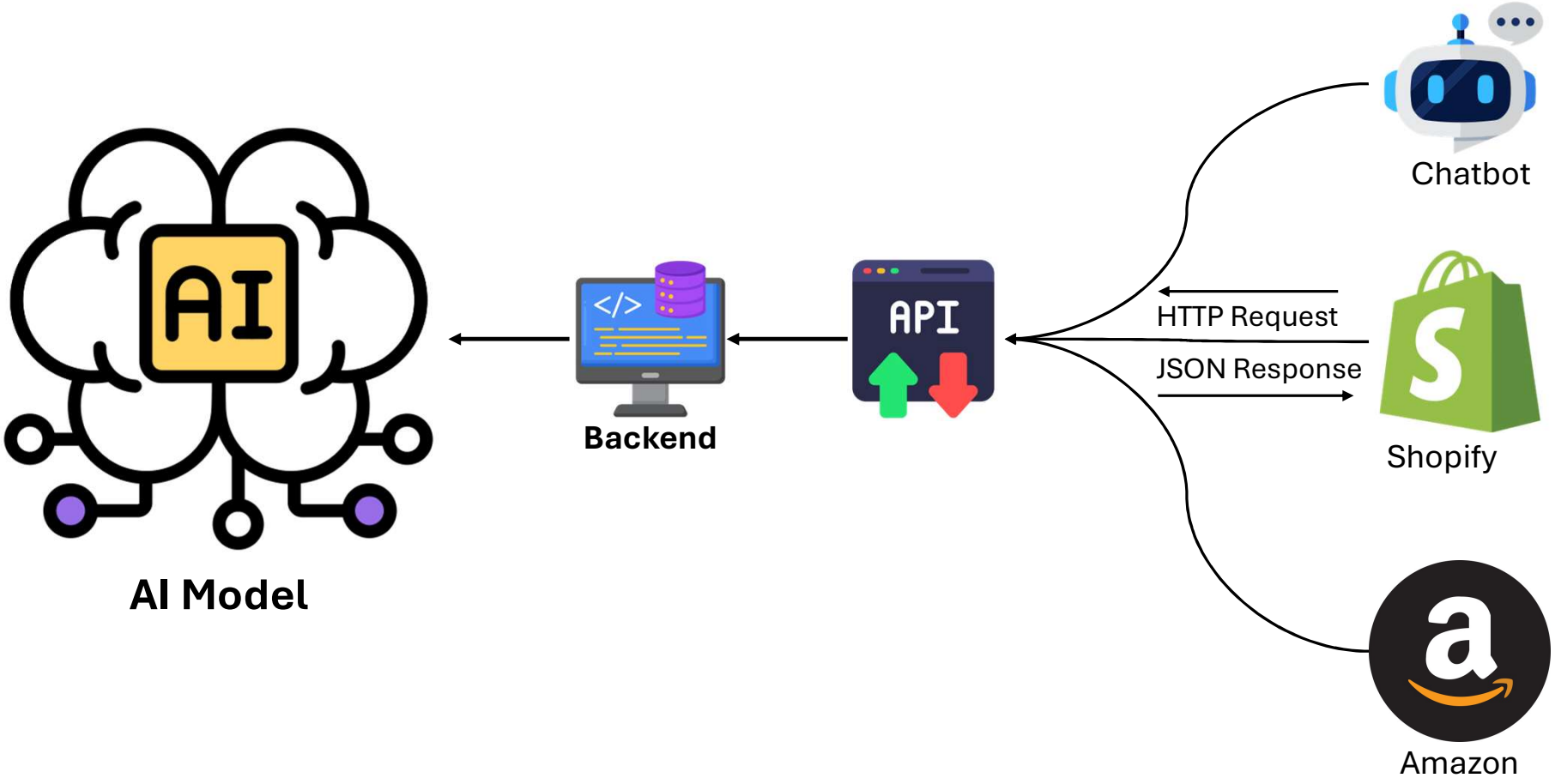


Social login APIs



Payment APIs

API – Machine Learning Perspective



API – Machine Learning Perspective

